Day 9 Study Session: Law of Total Probability and Partitioning Sample Spaces

Course: Actuarial Exam P
Module: General Probability (Phase 1)
Session: 1G.1 (SOA Objective 1g)

Learning Objectives

By the end of this session, you will be able to:

1. Define a **formal partition** of a sample space S using mutually exclusive and collectively exhaustive events.
2. State and apply the **Law of Total Probability (LTP)** to evaluate unconditional probabilities.
3. Translate actuarial multi-tier risk structures (e.g., policyholder risk classes, driver age groups, claim type categories) into precise LTP equations.
4. Solve multi-stage conditional probability problems using tree diagrams and partition tables.

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1. Core Theory: Partitions of a Sample Space

A collection of events $\{B_1, B_2, \ldots, B_k\}$ forms a **partition** of the sample space S if two conditions are met:

1. **Mutually Exclusive (Disjoint):** No two events share any outcomes.

$$B_i \cap B_j = \varnothing \quad \text{for all } i \neq j$$

2. **Collectively Exhaustive:** The union of all events covers the entire sample space S.

$$\bigcup_{i=1}^{k} B_i = B_1 \cup B_2 \cup \cdots \cup B_k = S$$

Because the partition elements cover S without overlap, their probabilities sum to 1:

$$\sum_{i=1}^{k} P(B_i) = P(B_1) + P(B_2) + \cdots + P(B_k) = 1$$

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2. The Law of Total Probability (LTP)

Let $\{B_1, B_2, \ldots, B_k\}$ be a partition of S such that $P(B_i) > 0$ for all i. For any event $A \subseteq S$, the unconditional probability $P(A)$ is obtained by summing the joint probabilities across all partition pieces:

$$P(A) = \sum_{i=1}^{k} P(A \cap B_i)$$

Applying the General Multiplication Rule ($P(A \cap B_i) = P(B_i) \cdot P(A \mid B_i)$):

$$P(A) = P(B_1)P(A \mid B_1) + P(B_2)P(A \mid B_2) + \cdots + P(B_k)P(A \mid B_k)$$

Two-Event Special Case

For a single partitioning event B and its complement B^c:

$$P(A) = P(B)P(A \mid B) + P(B^c)P(A \mid B^c)$$

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3. Concept Check Walkthroughs

Concept Check 1

Question: Let B be an event with $P(B) = 0.40$. If $P(A \mid B) = 0.70$ and $P(A \mid B^c) = 0.20$, calculate $P(A)$ using the Law of Total Probability.

Walkthrough:

1. **Identify Partition & Complement:**
   $P(B) = 0.40 \implies P(B^c) = 1 - 0.40 = 0.60$.
2. **Apply Two-Event LTP:**

$$P(A) = P(B)P(A \mid B) + P(B^c)P(A \mid B^c)$$

3. **Compute:**

$$P(A) = (0.40)(0.70) + (0.60)(0.20) = 0.28 + 0.12 = \mathbf{0.40}$$

Concept Check 2

Question: An insurer classifies property policies into Residential (60%) and Commercial (40%). The probability of a storm loss is 0.08 for Residential policies and 0.15 for Commercial policies. What is the total probability that a policy incurs a storm loss?

Walkthrough:

1. **Define Events:**
   R = Residential ($P(R) = 0.60$), C = Commercial ($P(C) = 0.40$).
   L = Storm Loss. $P(L \mid R) = 0.08$, $P(L \mid C) = 0.15$.
2. **Apply LTP:**

$$P(L) = P(R)P(L \mid R) + P(C)P(L \mid C)$$

3. **Compute:**

$$P(L) = (0.60)(0.08) + (0.40)(0.15) = 0.048 + 0.060 = \mathbf{0.108}\ (10.8\%)$$

Concept Check 3

Question: True or False? The events E_1, E_2, E_3 form a valid partition of sample space S if $P(E_1) = 0.40$, $P(E_2) = 0.35$, $P(E_3) = 0.25$, and $P(E_1 \cap E_2) = 0.05$.

Walkthrough:

1. **Check Requirements:** A partition requires events to be mutually exclusive ($P(E_i \cap E_j) = 0$ for $i \neq j$).
2. **Test:** Here, $P(E_1 \cap E_2) = 0.05 \neq 0$.
3. **Conclusion:** Because E_1 and E_2 overlap, they are not mutually exclusive.
   Answer: False

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4. Day 9 Practice Problem Set & Walkthroughs

Problem 1

Scenario: An insurance company categorizes its policyholders into Young (Y) and Mature (M). Young drivers make up 30% of policyholders ($P(Y) = 0.30$).

- $P(\text{Claim} \mid Y) = 0.20$
- $P(\text{Claim} \mid M) = 0.05$

Question: Calculate $P(C)$, the overall probability a randomly chosen policyholder files a claim.

Solution:

1. $P(M) = 1 - P(Y) = 1 - 0.30 = 0.70$.
2. Apply LTP:

$$P(C) = P(Y)P(C \mid Y) + P(M)P(C \mid M)$$

$$P(C) = (0.30)(0.20) + (0.70)(0.05) = 0.060 + 0.035 = \mathbf{0.095}\ (9.5\%)$$

Problem 2

Scenario: A property portfolio has three risk tiers: Low Risk (B_1, 50%), Moderate Risk (B_2, 35%), and High Risk (B_3, 15%).

- $P(L \mid B_1) = 0.02$
- $P(L \mid B_2) = 0.08$
- $P(L \mid B_3) = 0.24$

Question: Calculate $P(L)$, the overall probability a policy incurs a property loss.

Solution:

1. Verify partition: $0.50 + 0.35 + 0.15 = 1.00$.
2. Apply General LTP:

$$P(L) = P(B_1)P(L \mid B_1) + P(B_2)P(L \mid B_2) + P(B_3)P(L \mid B_3)$$

$$P(L) = (0.50)(0.02) + (0.35)(0.08) + (0.15)(0.24)$$

$$P(L) = 0.010 + 0.028 + 0.036 = \mathbf{0.074}\ (7.4\%)$$

Problem 3

Scenario: An actuary models disability claims:

1. $P(\text{Injury } I) = 0.12$.
2. $P(\text{Disability } D \mid I) = 0.25$.
3. $P(\text{Disability } D \mid I^c) = 0.03$.

Question: Calculate the unconditional probability of a disability claim ($P(D)$).

Solution:

1. $P(I^c) = 1 - 0.12 = 0.88$.
2. Apply Two-Event LTP:

$$P(D) = P(I)P(D \mid I) + P(I^c)P(D \mid I^c)$$

$$P(D) = (0.12)(0.25) + (0.88)(0.03)$$

$$P(D) = 0.030 + 0.0264 = \mathbf{0.0564}\ (5.64\%)$$

Problem 4

Scenario: Claims arrive from Center A (40%), Center B (35%), and Center C (25%).

- $P(\text{Error } E \mid A) = 0.05$
- $P(\text{Error } E \mid B) = 0.08$
- $P(\text{Error } E \mid C) = 0.12$

Questions:

1. Calculate $P(E)$, the overall probability of a claim having an administrative error.
2. Calculate $P(E^c)$, the probability a claim is error-free.

Solution:

1. **Part 1:**

$$P(E) = P(A)P(E \mid A) + P(B)P(E \mid B) + P(C)P(E \mid C)$$

$$P(E) = (0.40)(0.05) + (0.35)(0.08) + (0.25)(0.12)$$

$$P(E) = 0.020 + 0.028 + 0.030 = \mathbf{0.078}\ (7.8\%)$$

2. **Part 2:**

$$P(E^c) = 1 - P(E) = 1 - 0.078 = \mathbf{0.922}\ (92.2\%)$$

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5. Day 9 Formula Summary

Concept	Formula	Notes
Partition Conditions	$B_i \cap B_j = \varnothing, \bigcup B_i = S$	Mutually exclusive & collectively exhaustive
Law of Total Probability (General)	$P(A) = \sum_{i=1}^{k} P(B_i)P(A \mid B_i)$	Unconditional probability across k partition tiers
Law of Total Probability (2 Events)	$P(A) = P(B)P(A \mid B) + P(B^c)P(A \mid B^c)$	Standard binary partition using complement